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EXPERIENCE

Physics Wallah

Apr 2023 – Present

Software Engineer II

- Built Physics Wallah's web video platform from 0 to org-wide production, serving **2.7M+ DAU** across live classes, VOD, and OTT behind a unified interactive player layer.
- Designed a **multi-engine playback SDK** (Video.js DASH/EME, AWS Live, YouTube, hls.js) with flag-driven runtime routing across live/VOD formats; improved reliability via UUID-scoped attempt/outcome analytics and Conviva QoE.
- Owned **DRM delivery** (Widevine, FairPlay, ClearKey/Pallycon), including Safari service-worker segment auth, signed CDN cookies, and proactive license refresh; added client-side EME/codec probing to optimize upstream stream selection.
- Architected **real-time classroom transport** with redundant poll/chat delivery over HLS ID3 metadata, WebSockets, and MQTT; implemented Firebase-gated failover and modular in-player features (chat, polls, doubts, notes, ratings, leaderboards).
- Improved playback reliability through correlated analytics, Sentry error tracking with staged rollout, and manifest retry/recovery paths for long-lived live sessions.
- Shipped **Video Copilot** — in-player AI assistant with Azure Speech-to-Text, video-frame annotations, and SSE-streamed LLM responses; integrated SME doubt routing and session feedback loops.
- Led frontend architecture for the **subscription learning platform** and a **Module Federation** micro-frontend shell enabling independent team deployments with shared routing and SDKs.
- Designed subscription/checkout systems: plan management, couponing, QR payments, orchestration, and Unleash feature-flag rollouts.

Tech: React, TypeScript, Redux Toolkit, TanStack React Query, Video.js, hls.js, DRM/EME, WebSockets, MQTT, Azure Speech SDK, Firebase, Conviva, Sentry, Unleash, Workbox

Physics Wallah

April 2022 – Mar 2023

Software Engineer I

- OTT Admin Features Shipped: CMS (videos, playlists, subtitles), subscription plans, trial access, and onboarding — on an internal **90+ module** Angular/TypeScript platform.
- Built config-driven CRUD workflows using internal form/table libraries with REST APIs, pagination, search, dependent dropdowns, and modal-based edit flows; accelerated feature delivery for ops teams.
- Implemented RBAC with route guards and field-level permissions; delivered S3 bulk-upload workflows with client-side validation (WebP/PDF, file-size limits) and Sentry-backed production monitoring via Jenkins/ArgoCD.

Tech: Angular, TypeScript, RxJS, REST, RBAC, AWS S3

PROJECT

VisualJS

visualjs.in

- Built a privacy-first JavaScript learning platform with **80+ chapters**, a live code visualizer, and project tutorials.
- Engineered a client-side execution pipeline (Babel AST instrumentation, Web Worker sandbox) with real-time call stack, memory, and event loop views via React Flow.
- Shipped on Next.js 16 + React 19 with SEO across 100+ routes; **122** core modules, 100% client-side execution.
- Designed a pedagogy layer that detects common JS misconceptions (closures, hoisting, references)

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, Java, SQL
Frontend	React, Next.js, Redux Toolkit, RTK Query, React Query, Zustand, Module Federation, Rspack, Webpack, HTML, CSS, Sass, Tailwind CSS, Storybook
Streaming & Media	Video.js, hls.js, DASH/MPD, DRM, EME, ABR, Shaka Player
Real-Time Systems	WebSockets, MQTT, Socket.io
Databases	MySQL, MongoDB, Redis
Cloud & Tools	AWS, Docker, Git, Linux, Firebase, Conviva, Sentry, Unleash, Maestro
Architecture	Micro-Frontends, Distributed Systems, Event-Driven Architecture, System Design, Design Patterns, Scalability, Caching, Fault Tolerance

EDUCATION

B.Tech, Computer Science and Information Technology

2018 – 2022

Dronacharya Group of Institutions